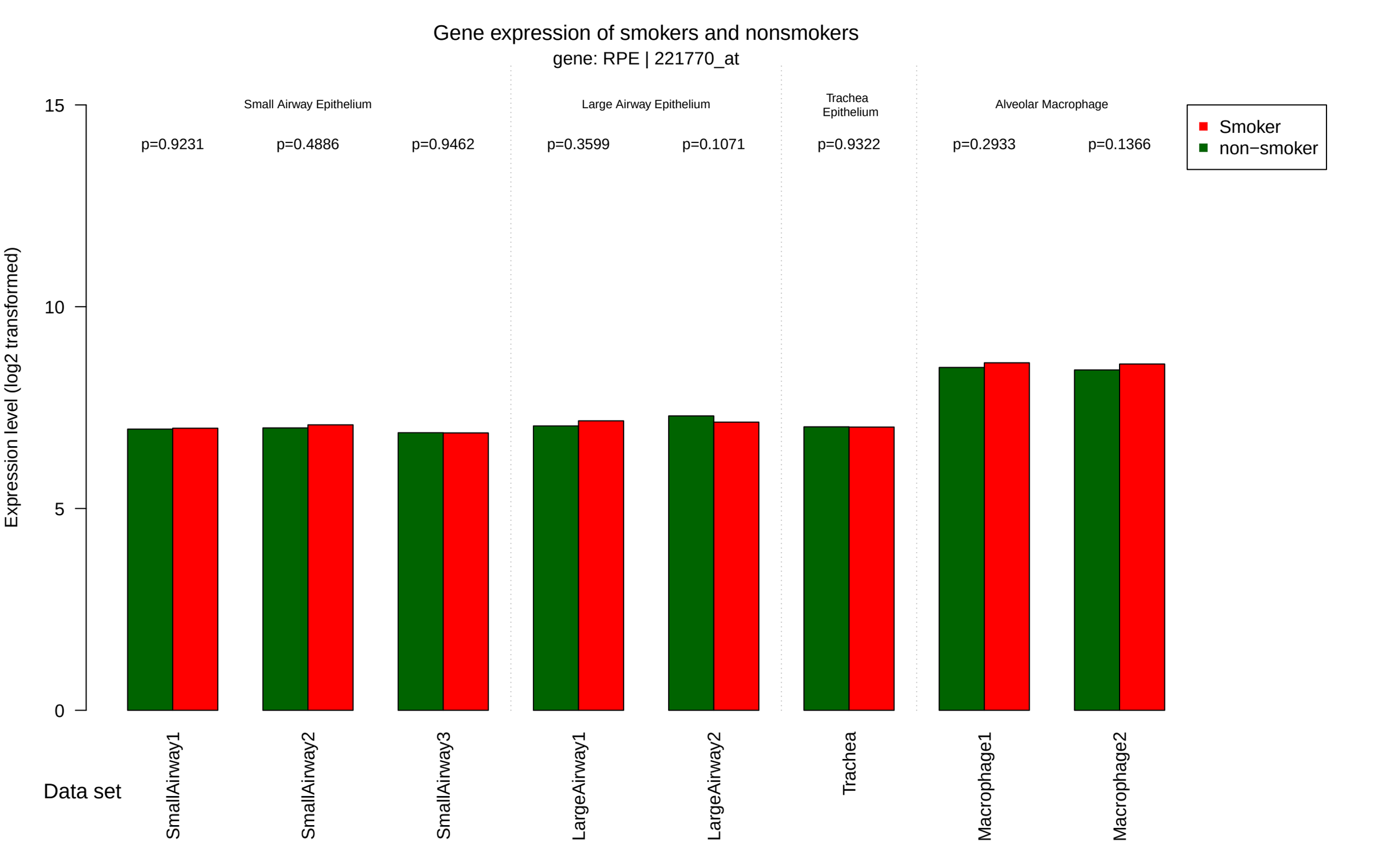
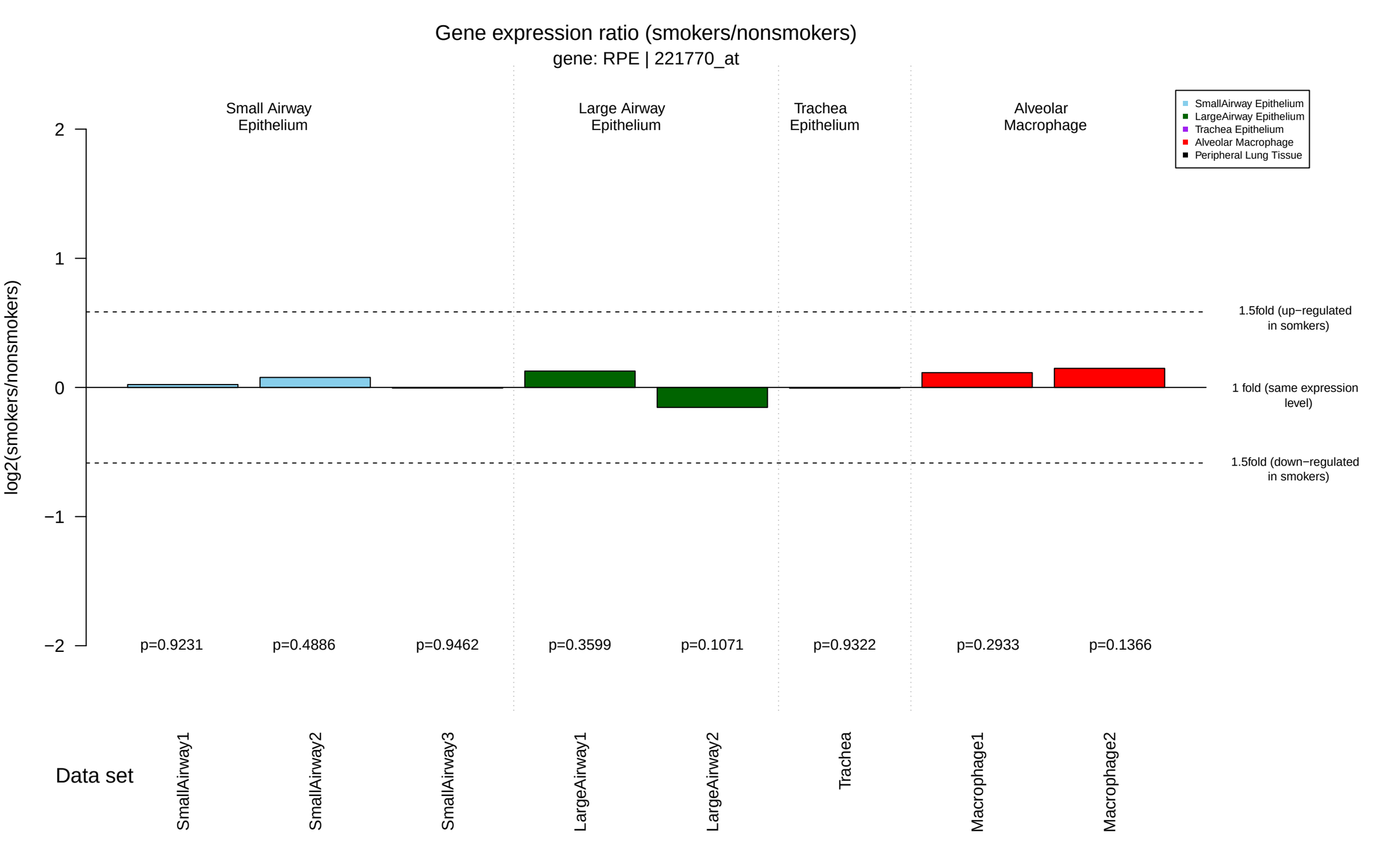
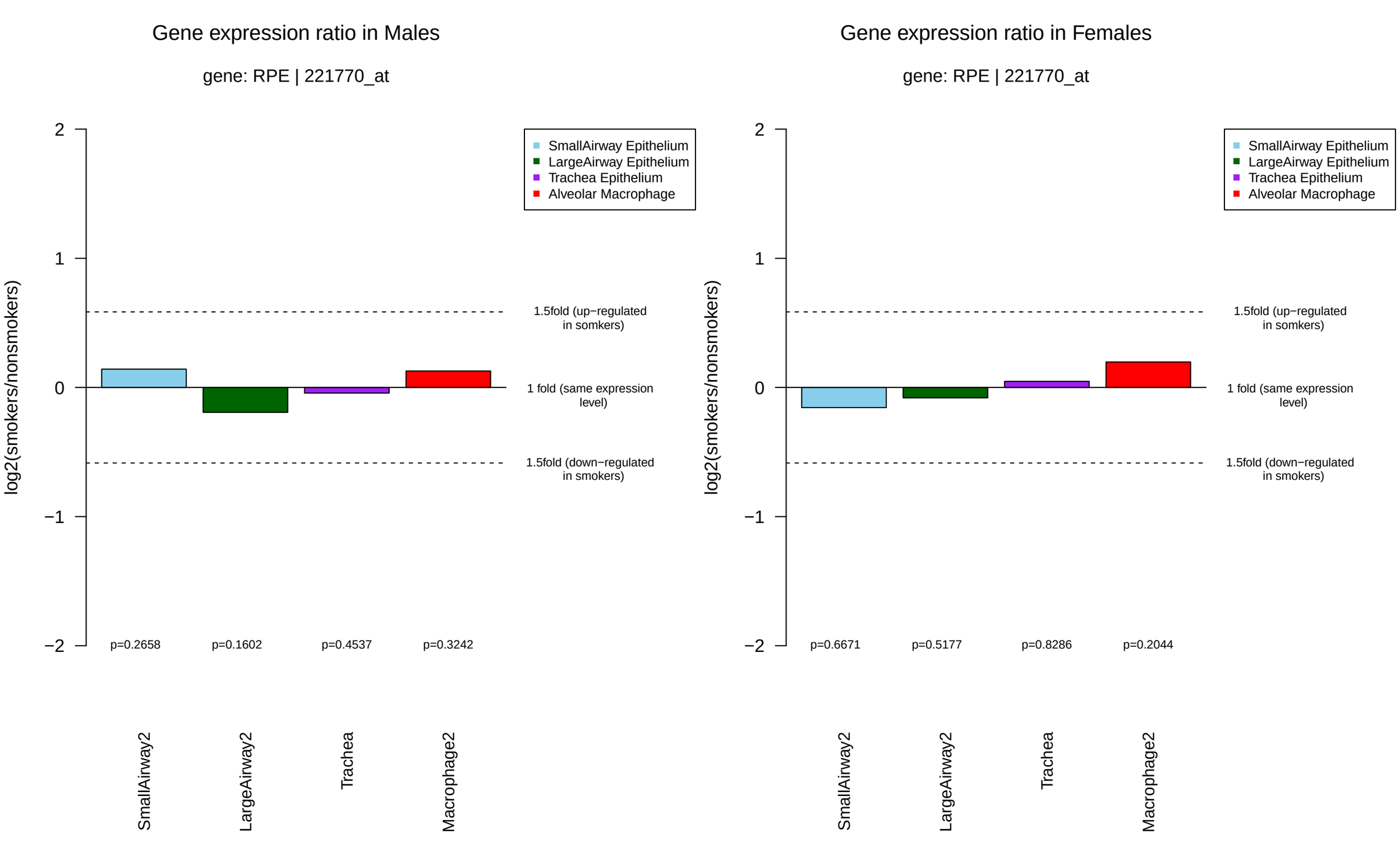
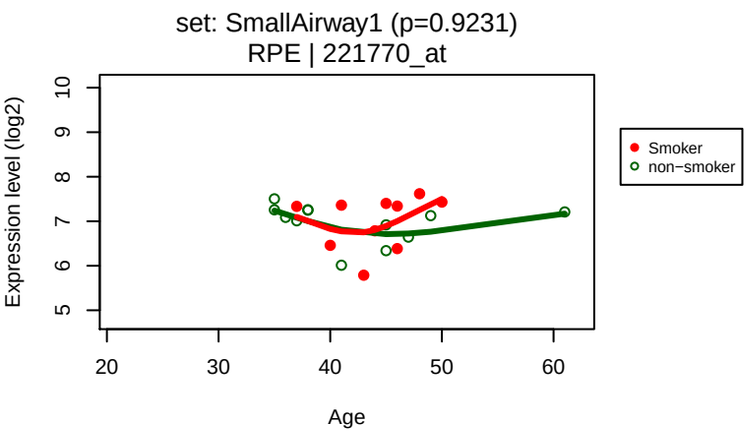
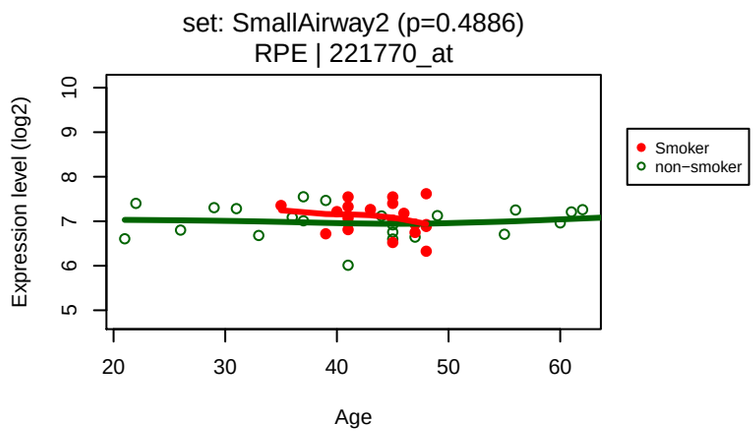
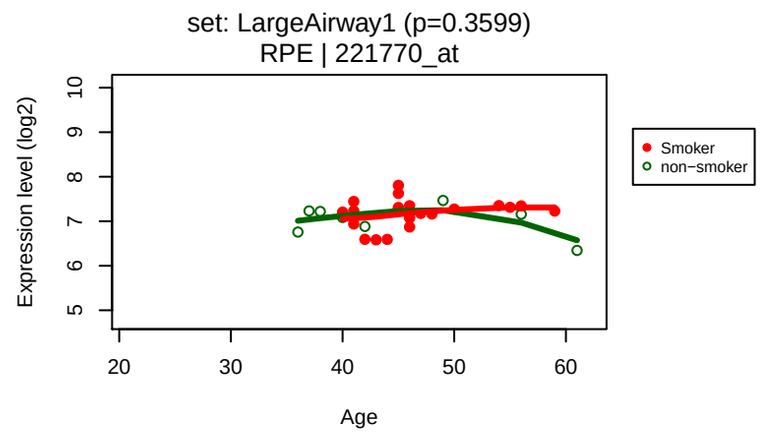

Gene expression of smokers and nonsmokers
gene: RPE | 221770_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9231
p=0.4886
p=0.9462
p=0.3599
p=0.1071
p=0.9322
p=0.2933
p=0.1366
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

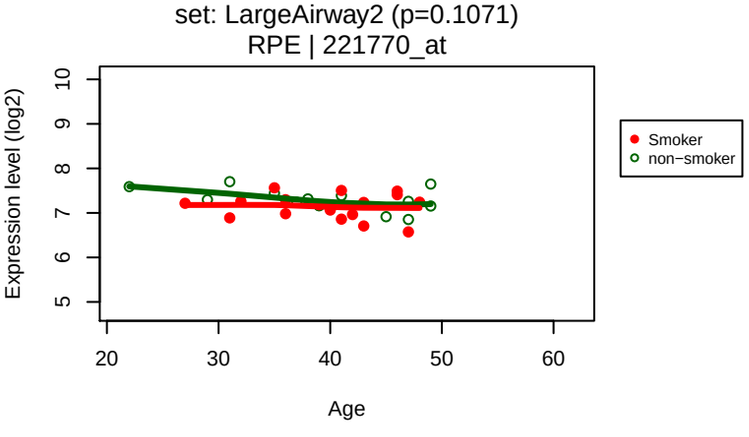

Gene expression ratio (smokers/nonsmokers)
gene: RPE | 221770_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9231
p=0.4886
p=0.9462
p=0.3599
p=0.1071
p=0.9322
p=0.2933
p=0.1366
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

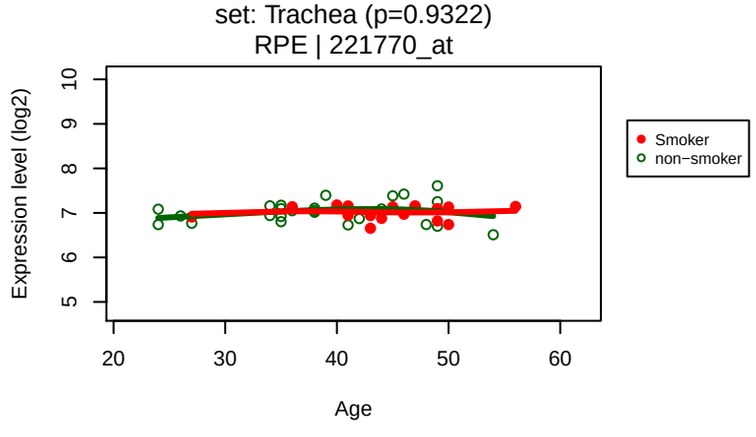

Gene expression ratio in Males
gene: RPE | 221770_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2658
p=0.1602
p=0.4537
p=0.3242
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RPE | 221770_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6671
p=0.5177
p=0.8286
p=0.2044
SmallAirway2
LargeAirway2
Trachea
Macrophage2

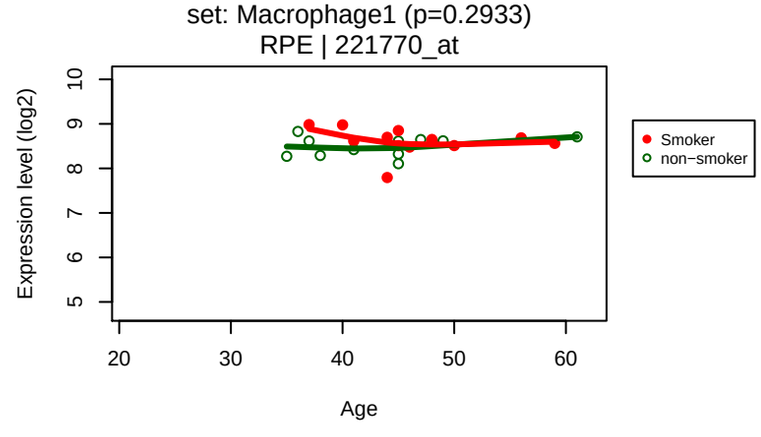

set: SmallAirway1 (p=0.9231)
RPE | 221770_at
Expression level (log2)
Age
Smoker
non−smoker

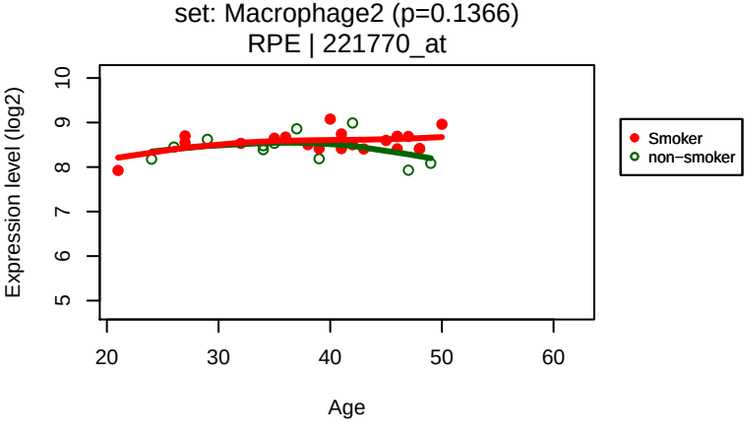

set: SmallAirway2 (p=0.4886)
RPE | 221770_at
Expression level (log2)
Age
Smoker
non−smoker

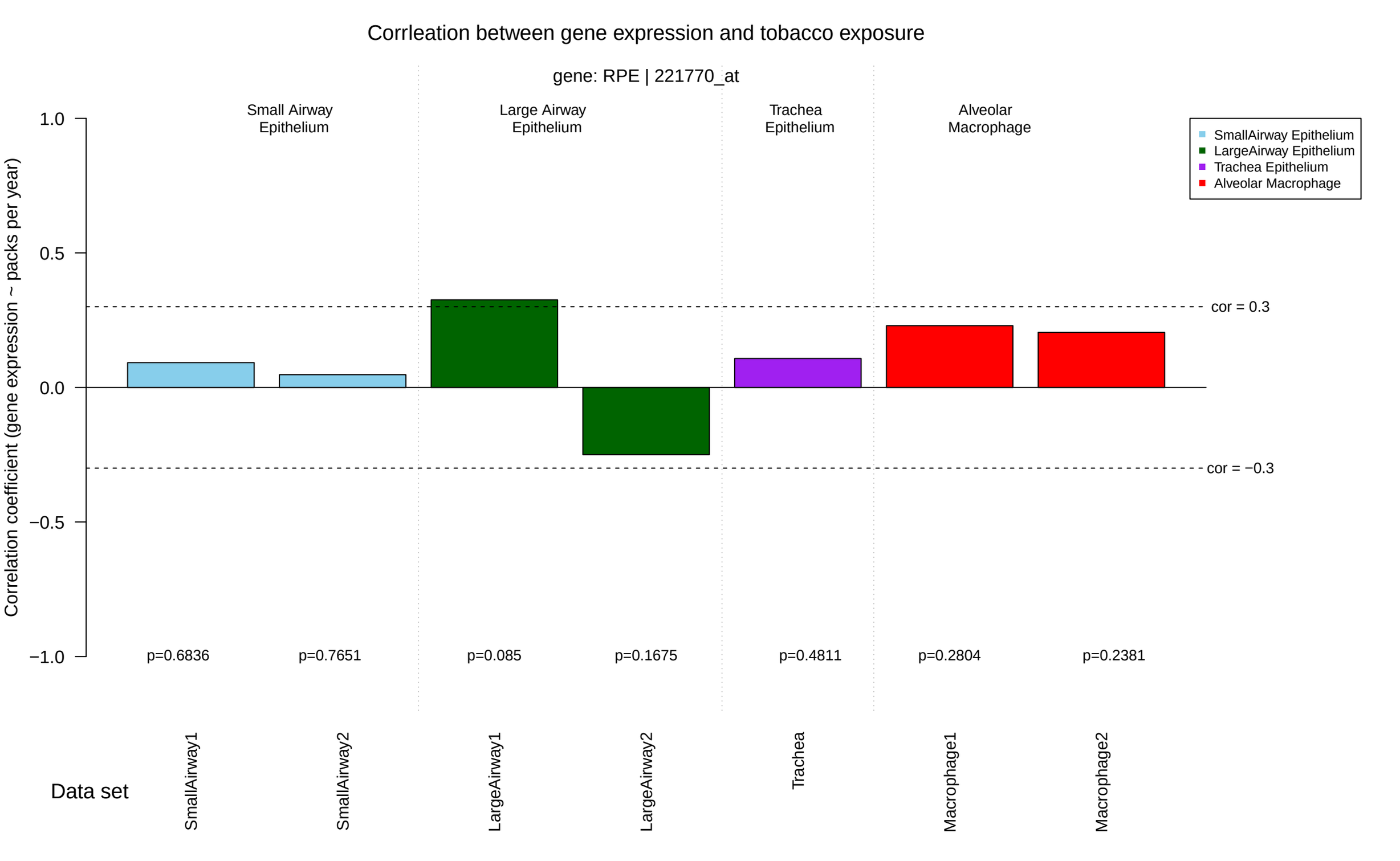

set: LargeAirway1 (p=0.3599)
RPE | 221770_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1071)
RPE | 221770_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9322)
RPE | 221770_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2933)
RPE | 221770_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1366)
RPE | 221770_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RPE | 221770_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6836
p=0.7651
p=0.085
p=0.1675
p=0.4811
p=0.2804
p=0.2381
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2